
Analysing the Effect of Periods of Economic Distress on Road Accidents in the UK

ECO-3402: Economic Forecasting and Analysis

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Abstract.

This paper examines the impact of macroeconomic determinants on traffic accidents in the UK from 1997 to 2024 through monthly data sets, which include 336 observations. The ARDL-ECM technique is utilized to estimate the dynamic linkages between total accidents and macroeconomic variables such as unemployment, inflation, GDP growth, and fuel prices, alongside controls for the number of registered vehicles and fuel usage. The stationarity of the series is analyzed using the Dickey-Fuller Test, while the Zivot-Andrews test is applied to detect endogenous structural shifts. Dummy variables for the 2008 Financial Crisis, the COVID-19 pandemic, and other economic shocks are included in the model. Long-run findings show that unemployment and inflation have the highest correlation with increasing traffic accidents. In contrast, the short-run analysis shows a suppression effect, implying that when costs increase, individuals become more cautious on roads. Additionally, GDP growth is insignificant in the long run. The 2008 financial crisis significantly increased accident rates independent of the macroeconomic variables, but the COVID-19 shock did not contribute to additional accidents because its impacts were subsumed by variable changes.

Contents

1	Introduction	1
2	Literature Review	1
3	Data Collection and Processing	3
3.1	Data Sources	3
3.2	Stationarity and Structural Breaks	4
4	Empirical Analysis and Results	6
4.1	Model Identification	6
4.2	Long-Run Anchoring	8
4.3	Short-Run Impacts	8
4.4	Effect of Structural Breaks	8
4.5	Error correction efficiency	9
5	Robustness Checks post Estimation	9
5.1	Diagnostic Tests	9
5.2	CUSUM Residual Plot	10
6	Policy Recommendations	11
6.1	Economic Intensity and the "High-Income" Safety Plateau	11
6.2	Macroeconomic Shocks and Structural Shifts in Risk	11
6.3	Decoupling Fuel Costs: The EV Transition Risk	11
7	Conclusion	11
8	Appendix	12
8.1	ARDL Model vs VAR Model	13
8.2	ARDL with level variables vs ARDL with first differenced variables	15
	References	16

1 Introduction

Worldwide, approximately 1.19 million people die in road accidents each year, with an additional 20–50 million individuals suffering non-fatal injuries. Road accidents impose significant costs on individuals, their families, and the broader economy. The World Health Organisation estimates that road accidents cost countries nearly 3% of their GDP, with developing countries bearing a disproportionate share of this burden. Although these countries account for roughly 60% of the world's vehicles, they account for nearly 92% of global road accidents¹. This underscores that road accidents are not only a public health concern but also a matter of substantial social and economic consequence.

While existing research often focuses on individual-level determinants of road accidents, such as weather conditions and driver behaviour, this paper examines whether accidents are influenced by macroeconomic conditions. In particular, it investigates the relationship between road accidents and key macroeconomic indicators, including GDP, unemployment, and inflation. The analysis uses a comprehensive time series dataset from the United Kingdom, comprising individual accident data from 1979 to 2024. We convert this to monthly data consisting of 336 data points from January 1997 to December 2024. Although the initial objective of this study was to examine road accidents in India, limitations in the availability and consistency of high-frequency data necessitated a shift in focus to the United Kingdom.

The empirical framework employs an Autoregressive Distributed Lag Error Correction Model (ARDL-ECM) to estimate both short-run dynamics and long-run relationships between road accidents and macroeconomic variables. The model includes unemployment, inflation, and diesel prices as key explanatory variables, along with controls such as new car registrations, diesel consumption, and petrol consumption. The ARDL approach is well suited to this setting, as it accommodates variables with different orders of integration while allowing for the estimation of both equilibrium relationships and short-run adjustments. Stationarity is assessed using the Dickey–Fuller test, and the Zivot–Andrews test is used to account for endogenous structural breaks. Accounting for such breaks is particularly important given the extent of macroeconomic volatility over the sample period.

The study utilises an ARDL(2,0,2,2,2,4,0,2) model to analyse the short- and long-term dynamics. The long-run results indicate that unemployment and inflation are the most substantial drivers of increased road accidents (increasing accident counts by 180.800 and 74.419 per unit increase, respectively), as well as significant but smaller long-term impacts from petrol prices and car registrations. Conversely, the short-run dynamics reveal an initial suppression of accidents from rising living costs and job losses, contrasted by temporary spikes in accidents during economic booms.

The error correction term (-1.441) is highly significant and confirms a strong long-run equilibrium. Additionally, while the 2008 Financial Crisis independently drove up accident rates beyond standard macroeconomic effects (an increase of 65.107), the impacts of the COVID-19 pandemic were largely absorbed by shifts in underlying variables like GDP and unemployment. Diagnostic and CUSUM tests confirm the model's strong predictive accuracy and structural stability over the sample period.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data and its construction. Sections 4 and 5 present the empirical analysis and robustness checks. Section 6 concludes with policy implications and key insights.

2 Literature Review

A well-established body of international literature demonstrates a significant and often counterintuitive relationship between economic prosperity and road traffic fatalities: as economies expand and employment rises, road deaths tend to increase. Partyka² quantifies this link, using 23 years

of U.S. data (1960–1982) alongside labour force statistics to build a linear regression model. The study found that a shift of workers from unemployed to employed was associated with an increase in fatalities, implying that economic activity directly amplifies road risk. Conversely, external shocks that depressed economic activity — the 1974 oil embargo and the introduction of the 55 mph national speed limit — were estimated to have reduced fatalities.

This procyclical pattern has been observed at the cross-national level. Gerdtham and Ruhm³, using annual data from 23 OECD countries over nearly four decades (1960–1997), applied fixed-effects regression to show that a decrease in the unemployment rate was associated with a growth in total mortality. The critical review by Hakim et al.⁴ further corroborated this pattern, surveying 15 macro-level models and concluding that unemployment is consistently found to be negatively related to accident rates with injuries, largely because reduced income constrains non-work-related travel.

Further evidence comes from Bougueroua and Carnis⁵, who applied a Vector Error Correction Model (VECM) to Algerian data spanning 1970–2013, examining the simultaneous short- and long-run relationships between GDP per capita, fuel consumption, and traffic accidents. Their results showed that GDP per capita positively and significantly influenced the number of traffic accidents in both the short and long run. Notably, the study also demonstrated that policy intervention through new traffic regulations introduced in 2009 reduced accident numbers while accounting for economic conditions, suggesting that policy can be effective in reducing traffic deaths independently of macroeconomic trends.

While the link between growth and fatalities is strong, it is not linear across all stages of development. Bishai et al.⁶, drawing on data from 41 countries between 1992 and 1996, confirmed an inverted-U relationship using fixed and random effects regression in a log-log framework. In lower-income countries, an increase in GDP per capita raised crashes, injuries, and deaths. However, in wealthier countries (above a GDP per capita threshold of approximately \$1,500–\$8,000), GDP growth was associated with a reduction in traffic deaths, though crashes and injuries did not decline proportionally. The authors attributed this divergence to improvements in trauma care and occupant protection in high-income settings rather than to fewer accidents occurring. This implies that developing countries are particularly vulnerable to road traffic incidents, a finding further substantiated by Kopits and Cropper⁷, who similarly documented an inverted-U relationship between per capita income and traffic fatality rates using cross-national panel data.

The Nigerian case, studied by Akinyemi⁸ using the Autoregressive Distributed Lag (ARDL) cointegration framework on 26 years of data (1991–2016), offers a developing-country perspective that tempers some of the above findings. In the long run, an increase in GDP was associated with a decrease in crashes and fatalities in Nigeria, but an increase in injuries. Notably, the unemployment rate had no statistically significant effect on road safety in Nigeria, a departure from OECD findings that may reflect informality in the labour market and the weak link between formal employment and vehicle use in that context.

Beyond identifying the economic drivers of road fatalities, a parallel strand of literature has focused on developing methodologically robust time series frameworks capable of capturing the temporal dynamics inherent in road safety data. Yannis et al.⁹ applied autoregressive non-linear models to fatality and vehicle ownership data from 16 European Union countries over 1970–2002, motivated by the observation that road safety data are serially correlated and therefore violate the assumptions of standard regression. Using an autoregressive log-transformed model, the study concluded that ignoring temporal serial correlation leads to underestimated standard errors and potentially invalid inference, reinforcing the case for explicitly autoregressive model structures when working with annual road safety time series.

Li et al.¹⁰ provide further methodological justification for the ARDL framework adopted in the present study. Applying an ARDL-ECM to Chinese road accident data, they examine the effects of employment, unemployment, highway mileage, and private vehicle ownership on road traffic outcomes. They justify the ARDL approach over alternative cointegration methods on the grounds that it imposes lower data requirements and performs reliably in small samples. Their findings are broadly consistent with the procyclical patterns documented in the broader literature, though some divergence emerges across different outcome measures.

Ghadi¹¹ applies the ARDL bounds-testing approach to monthly data from Jordan between 2012 and 2021, examining the relationship between tourism-related economic variables and three accident severity categories: fatalities, severe injuries, and minor injuries. Long-run cointegration was confirmed across all three models, with tourism income emerging as the only consistently significant explanatory variable. Notably, the effect of tourism income was asymmetric across time horizons: in the short run, higher tourism income was associated with a rise in minor accidents, while in the long run it was linked to reductions in both fatalities and severe injuries — an effect the author attributes to the road infrastructure investment that sustained tourism revenue enables. The study further validates the use of the ARDL approach for monthly road safety data.

Dadgar and Norström¹² address a subtly different question. Building on research which finds strong procyclical effects, they attempt to identify the net impact of GDP growth on traffic deaths rather than isolating long-run structural shifts from short-run cyclical variation. Using the WHO mortality database and GDP per capita for 18 OECD countries over 1960–2011, and employing an Error Correction Model with a weighted lag structure, they find that an increase in GDP per capita yields an immediate increase in traffic deaths, with the effect materialising contemporaneously rather than with a lag.

3 Data Collection and Processing

3.1 Data Sources

This paper relies on data from Road Safety Open Data provided by the UK Government¹³, alongside macroeconomic data from Haver Analytics¹⁴ and the RAC Foundation¹⁵. All variables are measured at monthly frequency from January 1997 to December 2024, yielding 336 observations over a 27-year period. While the UK Government provides highly granular accident-level data spanning several decades, the raw data were processed to obtain the total number of road accidents per month, including all motor vehicles.

The dependent variable in this study is the total number of road accidents per month (*TotalAccidents*). This is regressed on four primary macroeconomic variables: GDP, unemployment rate, inflation (*D.CPI*), and fuel prices. In addition, three control variables are included: new car registrations, petrol consumption, and diesel consumption per month, to account for changes in vehicle usage.

The inclusion of these controls is motivated by the relationship between economic activity and vehicle usage. As incomes rise, car ownership typically increases, which may mechanically raise the number of road accidents. While new car registrations provide a partial measure of the vehicle stock, they do not capture the total number of vehicles in circulation. Ideally, traffic volume or average distance travelled would be used; however, such data were not available at the required frequency. As a proxy, total petrol and diesel consumption are used to approximate vehicle usage. These variables help mitigate omitted variable bias arising from unobserved changes in traffic intensity.

Table 1: **Summary Statistics**

Variable	Obs	Mean	Std. Dev.	Min	Max
TotalAccidents	336	1,633.17	492.84	372.00	2,751.00
GDP	336	83.32	10.89	61.60	101.70
Unemployment	336	5.54	1.32	3.60	8.60
CPI	336	92.81	17.96	69.20	135.60
FuelPrices	336	111.35	30.23	60.22	197.38
Car_Reg	336	185,633.10	104,890.90	4,321.00	562,337.00
Tot_petrol	336	1,795.01	436.35	432.91	2,786.11
Tot_diesel	336	2,085.56	371.89	1,133.08	2,742.28

Note: Summary statistics are reported for the full sample of 336 monthly observations (1997–2024). CPI is first-differenced to construct the inflation variable (*inf*). *Car_Reg* denotes monthly new car registrations. *Tot_petrol* and *Tot_diesel* denote total petrol and diesel consumption per month respectively.

3.2 Stationarity and Structural Breaks

To assess stationarity, the Dickey–Fuller (DF) test was applied to all variables. The results indicate that all variables, except inflation, car registrations, and diesel consumption, exhibit Unit Roots. These variables were therefore first-differenced. Subsequent DF tests confirm that all transformed variables are stationary.

Table 2: **Augmented Dickey–Fuller Unit Root Test Results**

Variable	Test Statistic	1% CV	5% CV	10% CV	p-value
Δacc	−21.843	−3.453	−2.877	−2.570	0.0000***
Δgdp	−13.325	−3.453	−2.877	−2.570	0.0000***
$\Delta unemp$	−11.847	−3.453	−2.877	−2.570	0.0000***
<i>inf</i>	−15.829	−3.453	−2.877	−2.570	0.0000***
Δgas	−12.045	−3.453	−2.877	−2.570	0.0000***
<i>car_reg</i>	−25.025	−3.453	−2.877	−2.570	0.0000***
<i>tot_diesel</i>	−5.080	−3.453	−2.877	−2.570	0.0000***
$\Delta petrol$	−27.234	−3.453	−2.877	−2.570	0.0000***

Note: The table reports Augmented Dickey–Fuller (ADF) test statistics for each variable. The null hypothesis is that the series contains a unit root. All variables reject the null hypothesis at the 1% significance level, indicating stationarity. Critical values are MacKinnon approximate values. *** $p < 0.01$.

Given the long time horizon of the dataset, which includes multiple macroeconomic shocks, it is necessary to account for potential structural breaks. The Zivot–Andrews test is therefore employed to detect endogenous breaks in the level and/or trend of each series.

Table 3: **Zivot–Andrews Structural Break Test Results**

Variable	Lags	Min t-stat	Break Date	1% CV	5% CV	10% CV
Δacc	4	−12.534***	2020m8	−5.57	−5.08	−4.82
inf	3	−8.746***	2020m9	−5.57	−5.08	−4.82
Δgdp	4	−11.247***	2020m7	−5.57	−5.08	−4.82
Δgas	3	−8.136***	2020m7	−5.57	−5.08	−4.82
$\Delta unemp$	3	−9.219***	2011m10	−5.57	−5.08	−4.82
$\Delta petrol$	4	−12.862***	2020m7	−5.57	−5.08	−4.82
tot_diesel	3	−7.465***	2020m1	−5.57	−5.08	−4.82
car_reg	4	−16.378***	2020m4	−5.57	−5.08	−4.82

Note: Zivot–Andrews test allows for a structural break in both intercept and trend. The null hypothesis is that the series contains a unit root with no structural break. All variables reject the null at the 1% significance level. Break dates are identified endogenously at the minimum t-statistic. The clustering of break dates around 2020 reflects the COVID-19 shock; the exception is $\Delta unemp$, which breaks at 2011m10, corresponding to the post-GFC unemployment peak. *** $p < 0.01$.

The identified break dates align with major macroeconomic disruptions. Most notably, a cluster of breaks occurs during the COVID-19 period. For example, car registrations exhibit a structural break in April 2020, corresponding to national lockdown measures. Similarly, breaks in GDP, fuel consumption, and accidents between July and September 2020 reflect the resumption of economic activity following lockdown easing. Earlier breaks, such as the October 2011 unemployment shift, capture the delayed labour market effects of the Eurozone debt crisis and domestic austerity policies.

To account for these disruptions, four structural break dummy variables are introduced. A *COVID* dummy captures the period from April to September 2020. Additional dummies include *unemp_break* for October 2011, *fin_crisis* for September 2008, and a Brexit-related dummy (*brexit*) for June 2016. These controls help ensure that the estimated relationships are not driven by discrete structural shifts in the data.

We visualise the breaks in accordance to our dependent variable:

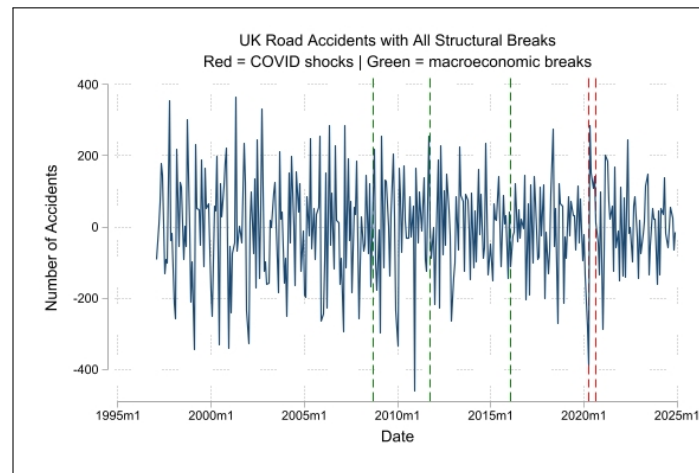


Figure 1: Structural Breaks

This preliminary analysis provides the foundation for the empirical strategy adopted in subsequent sections.

4 Empirical Analysis and Results

4.1 Model Identification

While trying to identify our model, we were divided between the Vector Autoregressive model {VAR} and the Autoregressive Distributed Lags model {ARDL}. Since the main aim of our study is to establish long run relationships between our macroeconomic indicators and road accidents, we chose the ARDL model, as its error correction framework captures this relationship better than VAR. Furthermore, the ARDL approach was used in our study because of its strength to deal with variables that exhibit different orders of integration, which in this study are the $I(1)$ economic shocks and $I(0)$ accident data. ARDL's ability to deal with lagged effects asymmetrically gives an added advantage in analyzing the delayed effect of economic shocks on road accidents.

We also encountered an obstacle of having the variables first differenced or not, in our model specification. Since ARDL accounts for an $I(1)$ integration, and that is the maximum level of integration we have, we thought to use all variables as level, regardless of their unit root. However, we ran our model with level variables and with first differenced variables and found a huge discrepancy between the values of R^2 for each model. The model that corrected for unit root wherever necessary provided better results than the one with variables at level. We further confirmed this performance using the Breusch-Godfrey test, IM White's test and the AIC and BIC criterion. All of these confirmed that our model with the first differenced variables is more accurate. The results for these are attached in the Appendix. Hence our model is defined as below. Post our estimation we understood which of the variables and their lags are significant in the long run, short run and we also found the error correction term. Hence in depicting our model, we have specified the same:

$$\begin{aligned}
 \Delta \text{Accidents}_t = & \alpha_0 + \underbrace{\phi \Delta \text{Acc}_{t-1}}_{\text{ADJ}} \\
 & + \underbrace{\beta_1 \text{TotDiesel}_t + \beta_2 \text{CarReg}_t + \beta_3 \inf_t + \beta_4 \Delta \text{GDP}_t + \beta_5 \Delta \text{Unemp}_t + \beta_6 \Delta \text{Gas}_t + \beta_7 \Delta \text{Petrol}_t}_{\text{LR}} \\
 & + \underbrace{\gamma_1 \Delta \text{Accidents}_{t-1}}_{\text{SR: Accidents}} \\
 & + \underbrace{\gamma_2 \Delta \text{CarReg}_t + \gamma_3 \Delta \text{CarReg}_{t-1}}_{\text{SR: Car Registrations}} \\
 & + \underbrace{\gamma_4 \Delta \inf_t + \gamma_5 \Delta \inf_{t-1}}_{\text{SR: Inflation}} \\
 & + \underbrace{\gamma_6 \Delta^2 \text{GDP}_t + \gamma_7 \Delta^2 \text{GDP}_{t-1}}_{\text{SR: GDP}} \\
 & + \underbrace{\gamma_8 \Delta^2 \text{Unemp}_t + \gamma_9 \Delta^2 \text{Unemp}_{t-1} + \gamma_{10} \Delta^2 \text{Unemp}_{t-2} + \gamma_{11} \Delta^2 \text{Unemp}_{t-3}}_{\text{SR: Unemployment}} \\
 & + \underbrace{\gamma_{12} \Delta^2 \text{Petrol}_t + \gamma_{13} \Delta^2 \text{Petrol}_{t-1}}_{\text{SR: Petrol}} \\
 & + \delta_1 \text{COVID}_t + \delta_2 \text{FinCrisis}_t + \varepsilon_t
 \end{aligned} \tag{1}$$

Post estimation we further specified our ARDL model into an ARDL(2,0,2,2,2,4,0,2) model. The results of our estimation are as follows:

Table 4: **ARDL(2,0,2,2,2,4,0,2) Estimation Results — Dependent Variable: $\Delta\text{accidents}$**

Component	Variable	Coeff.	Std. Err.	t	p-value
<i>Adjustment</i>					
	$\Delta\text{accidents}_{t-1}$	-1.441	0.073	-19.65	0.000***
<i>Long Run</i>					
	tot_diesel	-0.063	0.020	-3.19	0.002***
	car_reg	0.001	0.000	8.68	0.000***
	inf	74.419	16.601	4.48	0.000***
	Δgdp	3.745	6.019	0.62	0.534
	Δunemp	180.800	48.837	3.70	0.000***
	Δgas	-0.323	1.321	-0.24	0.807
	Δpetrol	0.486	0.098	4.94	0.000***
<i>Short Run</i>					
	$\Delta\text{accidents}_{t-1}$	0.123	0.046	2.67	0.008***
	$\Delta\text{car_reg}_t$	-0.001	0.000	-8.54	0.000***
	$\Delta\text{car_reg}_{t-1}$	-0.001	0.000	-9.06	0.000***
	Δinf_t	-99.367	20.199	-4.92	0.000***
	Δinf_{t-1}	-44.766	17.312	-2.59	0.010**
	$\Delta^2\text{gdp}_t$	30.501	7.150	4.27	0.000***
	$\Delta^2\text{gdp}_{t-1}$	15.627	5.870	2.66	0.008***
	$\Delta^2\text{unemp}_t$	-358.610	65.188	-5.50	0.000***
	$\Delta^2\text{unemp}_{t-1}$	-207.896	57.671	-3.60	0.000***
	$\Delta^2\text{unemp}_{t-2}$	-48.068	51.848	-0.93	0.355
	$\Delta^2\text{unemp}_{t-3}$	-175.940	44.625	-3.94	0.000***
	$\Delta^2\text{petrol}_t$	-0.523	0.109	-4.81	0.000***
	$\Delta^2\text{petrol}_{t-1}$	-0.181	0.057	-3.17	0.002***
<i>Structural Breaks</i>					
	covid	42.543	24.682	1.72	0.086*
	fin_crisis	65.107	20.149	3.23	0.001***
	Constant	-161.814	53.317	-3.03	0.003***
Observations		331			
R-squared		0.8114			
Adj. R-squared		0.7973			
Log Likelihood		-1977.841			
Root MSE		98.894			

Note: Sample: 1997m6–2024m12. Dependent variable is the first difference of monthly road accidents ($\Delta\text{accidents}$). Exogenous structural break dummies included for the COVID-19 shock and the Global Financial Crisis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.2 Long-Run Anchoring

Based on the long-run results, five explanatory variables are identified as statistically significant in explaining road accidents. Registrations of cars $\{\hat{\beta} = 0.0011, p < 0.001\}$, petrol consumption $\{\hat{\beta} = 0.486, p < 0.001\}$, inflation $\{\hat{\beta} = 74.419, p < 0.001\}$, unemployment $\{\hat{\beta} = 180.800, p < 0.001\}$ and diesel consumption $\{\hat{\beta} = -0.063, p < 0.001\}$ have strong impacts in the long run.

Total diesel consumption exhibits a negative long-run impact on road accidents $\{\hat{\beta} = -0.063, p = 0.002\}$, which means that an increase in the consumption of diesel fuel would lead to a decrease in the number of accidents. This finding can be attributed to the fact that most of the vehicles using diesel fuel are used for commercial purposes and the discipline among commercial drivers is higher than among private car drivers. On the other hand, GDP growth rates are not statistically significant in the long run $\{\hat{\beta} = 3.745, p = 0.534\}$, suggesting that individuals adapt well to this shock through time without any variations in the accident rate.

Positive long-term coefficient for unemployment supports the view that being unemployed for a long period of time results in financial hardships, deteriorating psychological wellbeing, and poor maintenance of vehicles, thereby leading to risky driving behavior. While inflation causes an immediate behavioral change, unemployment causes a gradually increasing pressure on individuals, eventually affecting road safety.

4.3 Short-Run Impacts

The short-run results indicate the presence of a variety of behavior responses across various regressors and lags. Inflation is shown to have a significant and powerful suppression effect on accidents $\{\hat{\gamma} = -99.367, p < 0.001 \text{ at } t; \hat{\gamma} = -44.766, p = 0.010 \text{ at } t - 1\}$, which seems to be consistent with the idea that households severely restrain their discretionary travel in the face of rising living costs. This effect is in contrast with the positive sign of the long-run coefficient of inflation, creating a classic error correction mechanism where the initial suppression of road activity is gradually replaced by behavioural adaptation over time. People will not stop driving forever and will eventually adapt to the higher prices.

Car registrations are also found to significantly suppress accidents in the short run $\{\hat{\gamma} = -0.001, p < 0.001 \text{ at both } t \text{ and } t - 1\}$, while capturing the initial increase in the road accident probability created by the exposure effect of more cars being on the road. The positive sign for GDP growth coefficients across two lags indicates that economic booms lead to rapid increases in road activity and number of accidents before any long-term effects become apparent $\{\hat{\gamma} = 30.501, p < 0.001 \text{ at } t; \hat{\gamma} = 15.627, p = 0.008 \text{ at } t - 1\}$.

There is a significant short-term suppression effect for unemployment with effects lasting for four periods $\{\hat{\gamma} = -358.610, p < 0.001\}$, while the contemporaneous effect reflects the initial drop in travel caused by the newly unemployed who no longer drive to work or engage in leisure travel. The effect of fuel prices is also statistically significant in the short term with respect to reducing the number of accidents $\{\hat{\gamma} = -0.523, p < 0.001 \text{ at } t; \hat{\gamma} = -0.181, p = 0.002 \text{ at } t - 1\}$, which is congruent with existing findings regarding the fuel elasticity of vehicle kilometres driven, in the transport economics literature.

4.4 Effect of Structural Breaks

Among the two dummy variables used to represent structural breaks in the final model, while the financial crisis dummy shows strong statistical significance $\{\hat{\delta} = 65.107, p = 0.001\}$, the COVID dummy is only statistically significant at the 10% level $\{\hat{\delta} = 42.543, p = 0.086\}$. The positive coefficient for the financial crisis dummy suggests that the period of the 2008 GFC had high accident counts beyond the effect of the macroeconomic determinants. This is supported by the findings that the austerity-induced reduction in the budget allocated to road maintenance, coupled with the

change in the composition of traffic in favour of risky travel patterns, had kept accident levels high despite contraction in the economy.¹⁶ The low statistical significance of the COVID dummy despite documented declines in traffic volumes due to lockdowns implies that most of the effects of the pandemic have been captured in the movement in macroeconomic variables such as unemployment and GDP growth in 2020. distances on unfamiliar roads, and increased stress on the drivers themselves.

4.5 Error correction efficiency

The adjustment coefficient of $-1.441\{p < 0.001\}$ supports the theoretical soundness of the ARDL error correction framework, providing evidence for the existence of long-run equilibrium among the variables, as evidenced by the Pesaran bounds test result, as shown in the robustness checks table in Section 5. The strength of the coefficient, having an absolute value greater than unity, suggests that there is slight overshooting of the long-run equilibrium by the system after experiencing any shock before settling down, which may be common in monthly accident data due to possible behavioural and enforcement reactions that might lead to over-adjustment. In economic terms, this suggests that adjustments to any deviations from the long-run equilibrium take place in about one to two months, indicating the high degree of sensitivity of road accident processes such as traffic enforcement and driving behaviour, as well as the use of vehicles, to any macroeconomic shocks.

5 Robustness Checks post Estimation

In order to validate the consistency of ARDL estimation results, four tests were performed. Serial correlation in the error terms was checked by using the Breusch-Godfrey LM test, while heteroscedasticity was examined through White's test. The suitability of the model was assessed based on Akaike and Bayesian Information Criteria (AIC and BIC). The Pesaran-Shin-Smith (2001) bounds test was conducted to confirm the existence of a long-run levels relationship among the variables. We also plotted the actual v fitted values {Via model estimation} to see if our model is strong. We get the following graph from that:

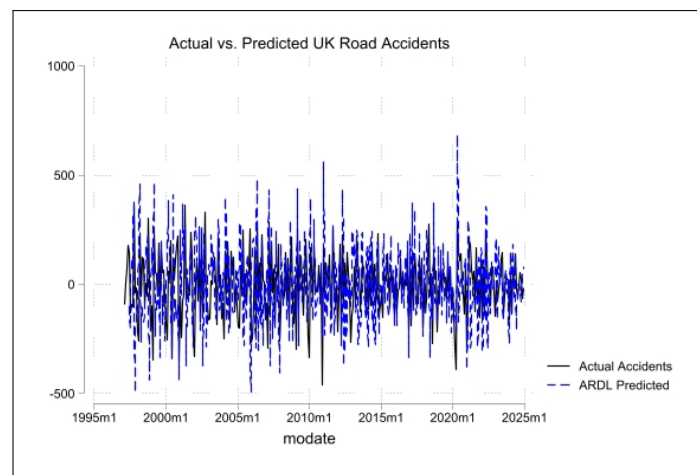


Figure 2: Pred v acc

We note that the ARDL model is estimated in first differences; therefore, the fitted value indicates changes in the number of accidents in the given month relative to the previous one rather than absolute levels of accidents. As seen in the figure, the model accurately represents both changes and trends in month-to-month fluctuations of actual data on road accidents during the whole sample period; in other words, the model shows similar oscillations as the differenced actual series.

5.1 Diagnostic Tests

The table below reflects the results of our post estimation tests aforementioned:

Table 5: **Diagnostic Tests**

Test	Component	Statistic	df	p-value
<i>Breusch–Godfrey LM Test for Autocorrelation</i>				
	4 lags	$\chi^2 = 8.640$	4	0.0707
<i>White’s Test for Heteroskedasticity</i>				
	Heteroskedasticity	$\chi^2 = 296.45$	295	0.4654
	Skewness	$\chi^2 = 22.63$	23	0.4827
	Kurtosis	$\chi^2 = 1.27$	1	0.2604
	Total (IM-test)	$\chi^2 = 320.34$	319	0.4684
<i>Information Criteria</i>				
	AIC	4003.681		$N = 331$
	BIC	4094.932		$df = 24$
<i>Pesaran–Shin–Smith (2001) Bounds Test</i>				
	F-statistic	63.484	> 4.26 (1% I(1) bound)	
	t-statistic	−19.652	< -5.19 (1% I(1) bound)	

Note: H_0 for Breusch–Godfrey: no serial correlation; fail to reject at 5% ($p = 0.071$). H_0 for White’s test: homoskedasticity; fail to reject ($p = 0.465$). For the Pesaran–Shin–Smith bounds test, H_0 is no long-run levels relationship; both the F-statistic (63.484) and t-statistic (−19.652) far exceed the 1% upper critical bounds of 4.26 and −5.19 respectively, confirming a stable long-run relationship. AIC and BIC reported from `estat ic`.

From the output, it can be observed that the chosen ARDL(2,0,2,2,2,4,0,2) model is sound from the diagnostic checks performed. From the LM statistic performed by Breusch–Godfrey, we fail to reject null hypothesis of serial correlation at 5% significance levels $\{\chi^2 = 8.640, p = 0.071\}$, implying that the lag length has been adequate for capturing temporal dependence in accidents.

White’s test indicates presence of homoskedasticity in all elements of IM test, implying that the variance of residuals is constant across the time period although there was substantial volatility in the economy from 1997 to 2024. The Pesaran–Shin–Smith (2001) bound test clearly confirms a stable long-term relationship between the variables, where the F-statistic value of 63.484 and t-statistic value of −19.652 both surpass the 1% upper critical bound of 4.26 and −5.19, respectively. As shown by model fit criteria values in terms of Akaike and Bayesian Information Criteria (AIC = 4003.681; BIC = 4094.932), the selected automatically lag structure of the model is optimal and there is no problem of over-fitting due to the minimal difference between the criteria.

5.2 CUSUM Residual Plot

We also run a CUSUM test on the residuals of the model, to confirm its stability. The CUSUM test confirms the stability of the parameters during the whole estimation period. The CUSUM statistic stays firmly inside the 5% critical limits throughout all 331 observations without reaching the red critical lines. Particularly, the statistic stays close to zero for most of the period and then falls slightly below zero around observations 150–250, roughly equivalent to the period of 2010–2020, before recovering to zero at the end of the period. In other words, although there is a slight negative trend during the middle part of the estimation period, it does not exceed the critical limit line and thus cannot be considered an indication of instability. Therefore, the coefficients of the ARDL(2,0,2,2,2,4,0,2) model are proven to be stable through the whole estimation period from 1997 to 2024, indicating that there is no structural break in the residuals other than what is captured by the COVID-19 and

financial crisis dummy variables.

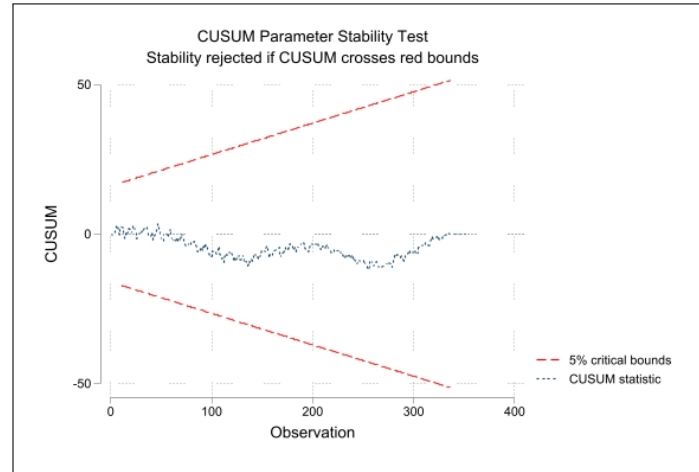


Figure 3: CUSUM Residual Plot

6 Policy Recommendations

6.1 Economic Intensity and the "High-Income" Safety Plateau

Our results offer a nuanced contrast to the literature on developing economies, such as the Nigerian case studied by Akinyemi⁸. While developing nations often see a reduction in fatalities as GDP growth enables basic safety infrastructure, this UK-based study demonstrates that in high-income settings, road safety is increasingly sensitive to the intensity of economic activity. In an environment where fundamental safety laws are already mature, the marginal fluctuations in accidents are dictated more by the procyclical volume of travel than by structural development. This suggests that for the UK, further gains in road safety may require alternative methods of monitoring and risk prevention.

6.2 Macroeconomic Shocks and Structural Shifts in Risk

This study aligns with Yannis et al.⁹ by showing that economic cycles can temporarily override technological gains in monitoring and safety. The strong statistical significance of the 2008 Financial Crisis dummy suggests that macroeconomic shocks do more than just change traffic volume; they potentially trigger shifts in "risk appetite" and infrastructure quality. For instance, the austerity-induced reductions in road maintenance budgets and changes in the composition of traffic toward higher-risk patterns can keep accident levels high even during an economic contraction. This underscores the need for focused intervention in infrastructure improvements and other prevention policies even during times of crises.

6.3 Decoupling Fuel Costs: The EV Transition Risk

An important finding of this research is the suppression effect of fuel prices and short-run inflation on accident rates. In economic terms, this reduces travel and commute on individual cars. This could either drive people to use more public transport or reduce travelling in general. However, as the UK transitions toward Electric Vehicles, it would be interesting to see if traffic growth and accidents end up increasing as this important link between fuel costs and traffic accidents reduces. Although, integration of newer technology like self-driving could work to mitigate this.

7 Conclusion

Through this paper, we investigated the dynamic relationship between macroeconomic fluctuations and road accidents in the United Kingdom from 1997 to 2024. Utilizing an Autoregressive Distributed Lag (ARDL) Error Correction Model, the analysis demonstrated that economic distress

and prosperity exert distinct effects on road safety. In the long run, persistent unemployment and inflation emerge as the primary drivers of increased accident rates, aggravated by higher petrol prices and vehicle registrations. Conversely, the short-run dynamics reveal a contrasting suppression effect where immediate economic pressures from inflation and job losses temporarily reduce road accidents.

Furthermore, the incorporation of structural breaks highlighted the risks posed by severe economic downturns. The 2008 Financial Crisis significantly elevated accident rates beyond standard macroeconomic effects, likely due to austerity measures impacting infrastructure maintenance and shifts toward riskier driving behaviours. In contrast, the shocks of the COVID-19 pandemic were largely captured by the extreme movements in underlying macroeconomic variables. Diagnostic and CUSUM tests confirmed the stability and strong predictive accuracy of the model, noting a rapid one-to-two-month adjustment back to equilibrium following economic shocks.

Ultimately, these findings underscore that in a high-income context like the UK, road safety remains highly sensitive to pro-cyclical travel volumes and macroeconomic volatility. As the nation navigates future economic uncertainties and the transition to electric vehicles, policymakers must adopt a proactive approach to ensure road safety is not compromised during periods of economic uncertainty.

8 Appendix

8.1 ARDL Model vs VAR Model

Table 6: ARDL and VAR Estimation Results — Dependent Variable: Δacc

Component	Variable	ARDL(2,0,2,2,2,4,0,2)		VAR(8)	
		Coeff.	p-value	Coeff.	p-value
<i>Adjustment / Own Lags</i>					
	Δacc_{t-1}	-1.441***	0.000	-0.455***	0.000
	Δacc_{t-2}	—	—	-0.310***	0.000
	Δacc_{t-3}	—	—	-0.299***	0.000
	Δacc_{t-4}	—	—	-0.319***	0.000
<i>Long Run (ARDL) / Selected Short-Run Lags (VAR)</i>					
	tot_diesel	-0.063***	0.002	-0.153	0.088
	car_reg	0.001***	0.000	0.001***	0.000
	inf	74.419***	0.000	43.499*	0.070
	Δgdp	3.745	0.534	5.625**	0.014
	$\Delta unemp$	180.800***	0.000	241.798***	0.000
	Δgas	-0.323	0.807	1.175	0.568
	$\Delta petrol$	0.486***	0.000	0.268***	0.003
<i>Short Run (ARDL Only)</i>					
	Δacc_{t-1}	0.123***	0.008	—	—
	Δcar_reg_t	-0.001***	0.000	—	—
	Δcar_reg_{t-1}	-0.001***	0.000	—	—
	Δinf_t	-99.367***	0.000	—	—
	Δinf_{t-1}	-44.766**	0.010	—	—
	$\Delta^2 gdp_t$	30.501***	0.000	—	—
	$\Delta^2 gdp_{t-1}$	15.627***	0.008	—	—
	$\Delta^2 unemp_t$	-358.610***	0.000	—	—
	$\Delta^2 unemp_{t-1}$	-207.896***	0.000	—	—
	$\Delta^2 unemp_{t-2}$	-48.068	0.355	—	—
	$\Delta^2 unemp_{t-3}$	-175.940***	0.000	—	—
	$\Delta^2 petrol_t$	-0.523***	0.000	—	—
	$\Delta^2 petrol_{t-1}$	-0.181***	0.002	—	—
<i>Structural Breaks</i>					
	covid	42.543*	0.086	30.889	0.245
	fin_crisis	65.107***	0.001	73.361***	0.002
	Constant	-161.814***	0.003	-71.720	0.281
Observations		331		327	
R-squared		0.8114		0.5794	
Adj. R-squared		0.7973		—	
AIC		4003.681		66.436	
Root MSE		98.894		103.842	

Note: The ARDL column reports long-run coefficients from the error correction form. The VAR column reports coefficients from the accidents equation only, showing the first lag of each variable. Both models estimated over the same variables with covid

8.2 ARDL with level variables vs ARDL with first differenced variables

Table 7: ARDL Estimation Results — First-Differenced vs. Level Variables

Component	Variable	ARDL — First Differenced ARDL(2,0,2,2,2,4,0,2)		ARDL — Levels ARDL(1,1,2,1,3,3,0,3)	
		Coeff.	p-value	Coeff.	p-value
<i>Adjustment</i>					
	Δacc_{t-1}	−1.441***	0.000	−1.291***	0.000
<i>Long Run</i>					
	tot_diesel	−0.063***	0.002	−0.191***	0.000
	car_reg	0.001***	0.000	0.001***	0.000
	inf / cpi	74.419***	0.000	32.492*	0.091
	gdp	3.745	0.534	3.714	0.144
	unemp	180.800***	0.000	8.679	0.264
	gas / fuelprices	−0.323	0.807	0.648	0.215
	petrol	0.486***	0.000	−0.023	0.614
<i>Short Run — Selected Coefficients</i>					
	Δacc_{t-1}	0.123***	0.008	—	—
	$\Delta \text{tot_diesel}_t$	—	—	0.240***	0.000
	$\Delta \text{car_reg}_t$	−0.001***	0.000	−0.001***	0.000
	$\Delta \text{car_reg}_{t-1}$	−0.001***	0.000	−0.001***	0.000
	Δinf_t	−99.367***	0.000	−66.853***	0.001
	Δgdp_t	30.501***	0.000	27.595***	0.000
	Δgdp_{t-1}	15.627***	0.008	−18.283***	0.002
	Δgdp_{t-2}	—	—	−22.389***	0.000
	Δunemp_t	−358.610***	0.000	−73.728	0.115
	$\Delta \text{unemp}_{t-1}$	−207.896***	0.000	118.630**	0.014
	$\Delta \text{unemp}_{t-2}$	−48.068	0.355	107.354**	0.016
	$\Delta \text{unemp}_{t-3}$	−175.940***	0.000	—	—
	Δpetrol_t	−0.523***	0.000	0.087	0.311
	$\Delta \text{petrol}_{t-1}$	−0.181***	0.002	0.404***	0.000
	$\Delta \text{petrol}_{t-2}$	—	—	0.239***	0.000
<i>Structural Breaks</i>					
	covid	42.543*	0.086	27.070	0.490
	fin_crisis	65.107***	0.001	23.010	0.534
	Constant	−161.814***	0.003	−332.678	0.213
Observations		331		331	
R-squared		0.8114		0.8062	
Adj. R-squared		0.7973		0.7917	
Log Likelihood		−1977.841		−1982.371	
Root MSE		98.894		100.257	

Note: Both models use Δacc as the dependent variable and are estimated over the same sample period (1997m6–2024m12) with `covid` and `fin_crisis` as exogenous structural break dummies. The first-differenced specification transforms I(1) macroeconomic variables (GDP, unemployment, fuel prices, petrol) into stationary first differences for cointegration. The levels specification uses the

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